KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5103B | 5 Credits: 4 | Program Elective | Source-grounded

Maintenance Engineering &

 Maintenance of printing machines is important for many reasons. The delay in

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5103B is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5103B.pdf from the uploaded official SITTTR syllabus archive.
Course code and title: preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Printing Technology
Course code	5103B
Course title	Maintenance Engineering &
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L:4, T:0, P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

18 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- Maintenance of printing machines is important for many reasons. The delay in
- production for an equipment failure can create serious problem because printing is a
- service industry.
- Like other technological fields, new concepts and applications are developing
- continuously in maintenance also. This proposed syllabus is based on latest changes.

Course outcomes

CO	Official outcome	Study level
CO1	15 Understanding precautions and lubrication. Demonstrate different drives used for power	See official syllabus
CO2	14 Understanding transmission. Demonstrate machine erection and reconditioning	See official syllabus
CO3	15 Understanding in detail also understands electrical elements.	See official syllabus
CO4	Illustrate Quality control in printing industry. 14 Understanding	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2
CO3	CO3 2

Row	Extracted official mapping
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	lubrication.	4	As specified
Module 2	Demonstrate different drives used for power transmission.	6	As specified
Module 3	understand electrical elements.	3	As specified
Module 4	Illustrate Quality control in printing industry.	5	As specified

MODULE 1

lubrication.

This module is presented from the official syllabus scope for Maintenance Engineering &. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: lubrication.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ Introduction to maintenance.

Introduction to maintenance. is an official topic in Module 1 of Maintenance Engineering &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Introduction to maintenance. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, introduction to maintenance. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Introduction to maintenance. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Introduction to maintenance. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Different types of maintenance

Different types of maintenance is an official topic in Module 1 of Maintenance Engineering &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Different types of maintenance using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, different types of maintenance should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Different types of maintenance means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Malayalam support: Module 1-□□ Different types of maintenance □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Safety precautions and housekeeping. 4 Remembering Types of lubrications and lubricating

Safety precautions and housekeeping. 4 Remembering Types of lubrications and lubricating is an official topic in Module 1 of Maintenance Engineering &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Safety precautions and housekeeping. 4 Remembering Types of lubrications and lubricating using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, safety precautions and housekeeping. 4 remembering types of lubrications and lubricating should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What Safety precautions and housekeeping. 4 Remembering Types of lubrications and lubricating means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Safety precautions and housekeeping. 4 Remembering Types of lubrications and lubricating □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

▼ 4 Understanding methods in printing industry Maintenance - Definition, Objectives and functions. Types of Maintenance - Planned maintenance and unplanned maintenance. Planned maintenance - Types-Preventive Maintenance, Predictive Maintenance, and Scheduled maintenance. Unplanned maintenance - Breakdown Maintenance. Contract maintenance. - Preventive Maintenance - Advantages, Functions - Planning, scheduling - Repair cycles, Dispatching and Controlling. Categories - Inspection, Small repair, Medium repair and Overhauling. Safety Precautions and House Keeping - safety precautions to be followed in press area - Aids to good housekeeping. Lubrication - Introduction - Advantages - Types of Lubricants - Solid, Semisolid and Liquid. Lubrication Schedule - Chart and Paint Marks. Methods of Lubrication - Gravity feed method-Wick feed, oil feed and Drip Feed , Force feed or Pressure feed method - Oil can, grease gun and grease cup, Splash method - Ring oiler, and Continuous/Multipoint flow method.

4 Understanding methods in printing industry Maintenance - Definition, Objectives and functions. Types of Maintenance - Planned maintenance and unplanned maintenance. Planned maintenance - Types-Preventive Maintenance, Predictive Maintenance, and Scheduled maintenance. Unplanned maintenance - Breakdown Maintenance. Contract maintenance. - Preventive Maintenance - Advantages, Functions - Planning, scheduling - Repair cycles, Dispatching and Controlling. Categories - Inspection, Small repair, Medium repair and Overhauling. Safety Precautions and House Keeping - safety precautions to be followed in press area - Aids to good housekeeping. Lubrication - Introduction - Advantages - Types of Lubricants - Solid, Semisolid and Liquid. Lubrication Schedule - Chart and Paint Marks. Methods of Lubrication - Gravity feed method-Wick feed, oil feed and Drip Feed , Force feed or Pressure feed method - Oil can, grease gun and grease cup, Splash method - Ring oiler, and Continuous/Multipoint flow method. is an official topic in Module 1 of Maintenance Engineering &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding methods in printing industry Maintenance - Definition, Objectives and functions. Types of Maintenance - Planned maintenance and unplanned maintenance. Planned maintenance - Types-Preventive Maintenance, Predictive Maintenance, and Scheduled maintenance. Unplanned maintenance - Breakdown Maintenance. Contract maintenance. - Preventive Maintenance - Advantages, Functions - Planning, scheduling - Repair cycles, Dispatching and Controlling. Categories - Inspection, Small repair, Medium repair and Overhauling. Safety Precautions and House Keeping - safety precautions to be followed in press area - Aids to good housekeeping. Lubrication - Introduction - Advantages - Types of Lubricants - Solid, Semisolid and Liquid. Lubrication Schedule - Chart and Paint Marks. Methods of Lubrication - Gravity feed method-Wick feed, oil feed and Drip Feed , Force feed or Pressure feed method - Oil can, grease gun and grease cup, Splash method - Ring oiler, and Continuous/Multipoint flow method. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding methods in printing industry maintenance - definition, objectives and functions. types of maintenance - planned maintenance and unplanned maintenance. planned maintenance - types-preventive maintenance, predictive maintenance, and scheduled maintenance. unplanned maintenance - breakdown maintenance. contract maintenance. - preventive maintenance - advantages, functions - planning, scheduling - repair cycles, dispatching and controlling. categories - inspection, small repair, medium repair and overhauling. safety precautions and house keeping - safety precautions to be followed in press area - aids to good housekeeping. lubrication - introduction - advantages - types of lubricants - solid, semisolid and liquid. lubrication schedule - chart and paint marks. methods of lubrication - gravity feed method-wick feed, oil feed and drip feed , force feed or pressure feed method - oil can, grease gun and grease cup, splash method - ring oiler, and continuous/multipoint flow method. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding methods in printing industry Maintenance - Definition, Objectives and functions. Types of Maintenance - Planned maintenance and unplanned maintenance. Planned maintenance - Types-Preventive Maintenance, Predictive Maintenance, and Scheduled maintenance. Unplanned maintenance - Breakdown Maintenance. Contract maintenance. - Preventive Maintenance - Advantages, Functions - Planning, scheduling - Repair cycles, Dispatching and Controlling. Categories - Inspection, Small repair, Medium repair and Overhauling. Safety Precautions and House Keeping - safety precautions to be followed in press area - Aids to good housekeeping. Lubrication - Introduction - Advantages - Types of Lubricants - Solid, Semisolid and Liquid. Lubrication Schedule - Chart and Paint Marks. Methods of Lubrication - Gravity feed method-Wick feed, oil feed and Drip Feed , Force feed or Pressure feed method - Oil can, grease gun and grease cup, Splash method - Ring oiler, and Continuous/Multipoint flow method. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 4 Understanding methods in printing industry Maintenance - Definition, Objectives and functions. Types of Maintenance - Planned maintenance and unplanned maintenance. Planned maintenance - Types-Preventive Maintenance, Predictive Maintenance, and Scheduled maintenance. Unplanned maintenance - Breakdown Maintenance. Contract maintenance. - Preventive Maintenance - Advantages, Functions - Planning, scheduling - Repair cycles, Dispatching and Controlling. Categories - Inspection, Small repair, Medium repair and Overhauling. Safety Precautions and House Keeping - safety precautions to be followed in press area - Aids to good housekeeping. Lubrication - Introduction - Advantages - Types of Lubricants - Solid, Semisolid and Liquid. Lubrication Schedule - Chart and Paint Marks. Methods of Lubrication - Gravity feed method-Wick feed, oil feed and Drip Feed , Force feed or Pressure feed method - Oil can, grease gun and grease cup, Splash method - Ring oiler, and Continuous/Multipoint flow method. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Demonstrate different drives used for power transmission.

This module is presented from the official syllabus scope for Maintenance Engineering &. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Demonstrate different drives used for power transmission.
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

▼ Chain drives.

Chain drives. is an official topic in Module 2 of Maintenance Engineering &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Chain drives. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, chain drives. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Chain drives. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Chain drives. □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

▼ Belt drives.

Belt drives. is an official topic in Module 2 of Maintenance Engineering &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Belt drives. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, belt drives. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Belt drives. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Belt drives. □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Gear drives.

Gear drives. is an official topic in Module 2 of Maintenance Engineering &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Gear drives. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, gear drives. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Gear drives. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Gear drives. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▾ Bearings.

Bearings. is an official topic in Module 2 of Maintenance Engineering &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Bearings. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, bearings. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Bearings. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Bearings. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Cam and followers.

Cam and followers. is an official topic in Module 2 of Maintenance Engineering &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Cam and followers. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, cam and followers. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Cam and followers. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Cam and followers. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Springs. 2 Understanding Chain Drives - Introduction, Types of Chains(Briefly) - Roller Chain, Silent Chain, Ewart Chain and Bead Chain - Merits and Demerits of Chain Drives. Belt Drives - Introduction, Types of Belts(Briefly) - Flat belt, Rope belt, Tooth Belt, V belt and Timing Belt- Merits and Demerits of Belt drives. Gear Drives - Introduction, Types of Gears(Briefly) - Spur gear, Helical gear, Bevel gear, Worm gears and Herringbone gear - Merits and Demerits of gear drives. Maintenance and Lubrication of Drive Systems - Chain Drive, Belt Drive, and Gear Drive. Direct drive technology - Introduction - Advantages - Application in the printing field. Bearings - Introduction, Merits and Demerits. Types of Bearings (Briefly) - Sliding bearings and Antifriction bearings - Ball bearings, Needle bearings and Roller bearing. Cams and Follower - Advantages of cam and Follower. Types of Cams and Followers (Briefly) - Disk Cam, Translation Cam, Groove Plate Cam, Cylindrical Cam, Eccentric Cam and Tow Wipe Cam. Springs - Types of springs (Briefly) - Helical Spring, Conical spring, Volute Spring and Torsion Springs and its application. Demonstrate machine erection and reconditioning in detail also

Springs. 2 Understanding Chain Drives - Introduction, Types of Chains(Briefly) - Roller Chain, Silent Chain, Ewart Chain and Bead Chain - Merits and Demerits of Chain Drives. Belt Drives - Introduction, Types of Belts(Briefly) - Flat belt, Rope belt, Tooth Belt, V belt and Timing Belt- Merits and Demerits of Belt drives. Gear Drives - Introduction, Types of Gears(Briefly) - Spur gear, Helical gear, Bevel gear, Worm gears and Herringbone gear - Merits and Demerits of gear drives. Maintenance and Lubrication of Drive Systems - Chain Drive, Belt Drive, and Gear Drive. Direct drive technology - Introduction - Advantages - Application in the printing field. Bearings - Introduction, Merits and Demerits. Types of Bearings (Briefly) - Sliding bearings and Antifriction bearings - Ball bearings, Needle bearings and Roller bearing. Cams and Follower - Advantages of cam and Follower. Types of Cams and Followers (Briefly) - Disk Cam, Translation Cam, Groove Plate Cam, Cylindrical Cam, Eccentric Cam and Tow Wipe Cam. Springs - Types of springs (Briefly) - Helical

Spring, Conical spring, Volute Spring and Torsion Springs and its application. Demonstrate machine erection and reconditioning in detail also is an official topic in Module 2 of Maintenance Engineering &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Springs. 2 Understanding Chain Drives - Introduction, Types of Chains(Briefly) - Roller Chain, Silent Chain, Ewart Chain and Bead Chain - Merits and Demerits of Chain Drives. Belt Drives - Introduction, Types of Belts(Briefly) - Flat belt, Rope belt, Tooth Belt, V belt and Timing Belt- Merits and Demerits of Belt drives. Gear Drives - Introduction, Types of Gears(Briefly) - Spur gear, Helical gear, Bevel gear, Worm gears and Herringbone gear - Merits and Demerits of gear drives. Maintenance and Lubrication of Drive Systems - Chain Drive, Belt Drive, and Gear Drive. Direct drive technology - Introduction - Advantages - Application in the printing field. Bearings - Introduction, Merits and Demerits. Types of Bearings (Briefly) - Sliding bearings and Antifriction bearings - Ball bearings, Needle bearings and Roller bearing. Cams and Follower - Advantages of cam and Follower. Types of Cams and Followers (Briefly) - Disk Cam, Translation Cam, Groove Plate Cam, Cylindrical Cam, Eccentric Cam and Tow Wipe Cam. Springs - Types of springs (Briefly) - Helical Spring, Conical spring, Volute Spring and Torsion Springs and its application. Demonstrate machine erection and reconditioning in detail also using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, springs. 2 understanding chain drives - introduction, types of chains(briefly) - roller chain, silent chain, ewart chain and bead chain - merits and demerits of chain drives. belt drives - introduction, types of belts(briefly) - flat belt, rope belt, tooth belt, v belt and timing belt- merits and demerits of belt drives. gear drives - introduction, types of gears(briefly) - spur gear, helical gear, bevel gear, worm gears and herringbone gear - merits and demerits of gear drives. maintenance and lubrication of drive systems - chain drive, belt drive, and gear drive. direct drive technology - introduction - advantages - application in the printing field. bearings - introduction, merits and demerits. types of bearings (briefly) - sliding bearings and antifriction bearings - ball bearings, needle bearings and roller bearing. cams and follower - advantages of cam and follower. types of cams and followers (briefly) - disk cam, translation cam, groove plate cam, cylindrical cam, eccentric cam and tow wipe cam. springs - types of springs (briefly) - helical spring, conical spring, volute spring and torsion springs and its application. demonstrate machine erection and reconditioning in detail also should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Springs. 2 Understanding Chain Drives - Introduction, Types of Chains(Briefly) - Roller Chain, Silent Chain, Ewart Chain and Bead Chain - Merits and Demerits of Chain Drives. Belt Drives - Introduction, Types of Belts(Briefly) - Flat belt, Rope belt, Tooth Belt, V belt and Timing Belt- Merits and Demerits of Belt drives. Gear Drives - Introduction, Types of Gears(Briefly) - Spur gear, Helical gear, Bevel gear, Worm gears and Herringbone gear - Merits and Demerits of gear drives. Maintenance and Lubrication of Drive Systems - Chain Drive, Belt Drive, and Gear Drive. Direct drive technology - Introduction - Advantages - Application in the printing field. Bearings - Introduction, Merits and Demerits. Types of Bearings (Briefly) - Sliding bearings and Antifriction bearings - Ball bearings, Needle bearings and Roller bearing. Cams and Follower - Advantages of cam and Follower. Types of Cams and Followers (Briefly) -

Aspect	What to prepare
	Disk Cam, Translation Cam, Groove Plate Cam, Cylindrical Cam, Eccentric Cam and Tow Wipe Cam. Springs - Types of springs (Briefly) - Helical Spring, Conical spring, Volute Spring and Torsion Springs and its application. Demonstrate machine erection and reconditioning in detail also means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Springs. 2 Understanding Chain Drives - Introduction, Types of Chains(Briefly) - Roller Chain, Silent Chain, Ewart Chain and Bead Chain - Merits and Demerits of Chain Drives. Belt Drives - Introduction, Types of Belts(Briefly) - Flat belt, Rope belt, Tooth Belt, V belt and Timing Belt- Merits and Demerits of Belt drives. Gear Drives - Introduction, Types of Gears(Briefly) - Spur gear, Helical gear, Bevel gear, Worm gears and Herringbone gear - Merits and Demerits of gear drives. Maintenance and Lubrication of Drive Systems - Chain Drive, Belt Drive, and Gear Drive. Direct drive technology - Introduction - Advantages - Application in the printing field. Bearings - Introduction, Merits and Demerits. Types of Bearings (Briefly) - Sliding bearings and Antifriction bearings - Ball bearings, Needle bearings and Roller bearing. Cams and Follower - Advantages of cam and Follower. Types of Cams and Followers (Briefly) - Disk Cam, Translation Cam, Groove Plate Cam, Cylindrical Cam, Eccentric Cam and Tow Wipe Cam. Springs - Types of springs (Briefly) - Helical Spring, Conical spring, Volute Spring and Torsion Springs and its application. Demonstrate machine erection and reconditioning in detail also □□□□□□□□□□□□ □□□□□

definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

understand electrical elements.

This module is presented from the official syllabus scope for Maintenance Engineering &. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: understand electrical elements.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ Machine erection.

Machine erection. is an official topic in Module 3 of Maintenance Engineering &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Machine erection. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, machine erection. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Machine erection. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Machine erection. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▾ Machine reconditioning.

Machine reconditioning. is an official topic in Module 3 of Maintenance Engineering &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Machine reconditioning. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, machine reconditioning. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Machine reconditioning. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Machine reconditioning. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▾ Electrical drives. 5 Understanding Machine Erection and Reconditioning - Introduction, Equipments and Tools used for it, and application. Machine Erection - Equipment needed for erection - selection of location and environmental conditions - erection procedure for various prepress. Printing and finishing equipments and machinery. Reconditioning - Principles of reconditioning-repair methods of various parts - roller copperising and rerubberising - ebonite covering- dampening and inking systems - paper transport systems - cylinder bearing supports - eccentrics. Maintenance of electrical systems - AC motors and DC motors- electromagnetic friction couplings. Electromagnets - magnetic starters and contractors - limit switches - knife switches - micro switches starting and regulating rheostat - electric panels - electrical apparatus and electric wiring on the machines.

Electrical drives. 5 Understanding Machine Erection and Reconditioning - Introduction, Equipments and Tools used for it, and application. Machine Erection - Equipment needed for erection - selection of location and environmental conditions - erection procedure for various prepress. Printing and finishing equipments and machinery. Reconditioning - Principles of reconditioning-repair methods of various parts - roller copperising and rerubberising - ebonite covering- dampening and inking systems - paper transport systems - cylinder bearing supports - eccentrics. Maintenance of electrical systems - AC motors and DC motors- electromagnetic friction couplings. Electromagnets - magnetic starters and contractors - limit switches - knife switches - micro switches starting and regulating rheostat - electric panels - electrical apparatus and electric wiring on the machines. is an official topic in Module 3 of Maintenance Engineering &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Electrical drives. 5 Understanding Machine Erection and Reconditioning - Introduction, Equipments and Tools used for it, and application. Machine Erection - Equipment needed for erection - selection of location and environmental conditions - erection procedure for various prepress. Printing and finishing equipments and machinery. Reconditioning - Principles of reconditioning-repair methods of various parts - roller copperising and rerubberising - ebonite covering- dampening and inking systems - paper transport systems - cylinder bearing supports - eccentrics. Maintenance of electrical systems - AC motors and DC motors- electromagnetic friction couplings. Electromagnets - magnetic starters and contractors - limit switches - knife switches - micro switches starting and regulating rheostat - electric panels - electrical apparatus and electric wiring on the machines. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, electrical drives. 5 understanding machine erection and reconditioning - introduction, equipments and tools used for it, and application. machine erection - equipment needed for erection - selection of location and environmental conditions - erection procedure for various prepress. printing and finishing equipments and machinery. reconditioning - principles of reconditioning-repair methods of various parts - roller copperising and rerubberising - ebonite covering- dampening and inking systems - paper transport systems - cylinder bearing supports - eccentrics. maintenance of electrical systems - ac motors and dc motors- electromagnetic friction couplings. electromagnets - magnetic starters and contractors - limit switches - knife switches - micro switches starting and regulating rheostat - electric panels - electrical apparatus and electric wiring on the machines. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Electrical drives. 5 Understanding Machine Erection and Reconditioning - Introduction, Equipments and Tools used for it, and application. Machine Erection - Equipment needed for erection - selection of location and environmental conditions - erection procedure for various prepress. Printing and finishing equipments and machinery. Reconditioning - Principles of reconditioning-repair methods of various parts - roller copperising and rerubberising - ebonite covering-dampening and inking systems - paper transport systems - cylinder bearing supports - eccentrics. Maintenance of electrical systems - AC motors and DC motors- electromagnetic friction couplings. Electromagnets - magnetic starters and contractors - limit switches - knife switches - micro switches starting and regulating rheostat - electric panels - electrical apparatus and electric wiring on the machines. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Electrical drives. 5 Understanding Machine Erection and Reconditioning - Introduction, Equipments and Tools used for it, and application. Machine Erection - Equipment needed for erection - selection of location and environmental conditions - erection procedure for various prepress. Printing and finishing equipments and machinery. Reconditioning - Principles of reconditioning-repair methods of various parts - roller copperising and rerubberising - ebonite covering- dampening and inking systems - paper transport systems - cylinder bearing supports - eccentrics. Maintenance of electrical systems - AC motors and DC motors- electromagnetic friction couplings. Electromagnets - magnetic starters and contractors - limit switches - knife switches - micro switches starting and regulating rheostat - electric panels - electrical apparatus and electric wiring on the machines. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Illustrate Quality control in printing industry.

This module is presented from the official syllabus scope for Maintenance Engineering &. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Illustrate Quality control in printing industry.
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

▼ Quality. 1 Understanding Quality & Process control targets /

Quality. 1 Understanding Quality & Process control targets / is an official topic in Module 4 of Maintenance Engineering &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Quality. 1 Understanding Quality & Process control targets / using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, quality. 1 understanding quality & process control targets / should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Quality. 1 Understanding Quality & Process control targets / means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Quality. 1 Understanding Quality & Process control targets / □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ 4 Understanding patches. Electrical system, pneumatic system

4 Understanding patches. Electrical system, pneumatic system is an official topic in Module 4 of Maintenance Engineering &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding patches. Electrical system, pneumatic system using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding patches. electrical system, pneumatic system should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 4 Understanding patches. Electrical system, pneumatic system means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 4 Understanding patches. Electrical system, pneumatic system □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▾ 3 Understanding and hydraulic system maintenance. Quality/process standardization

3 Understanding and hydraulic system maintenance. Quality/process standardization is an official topic in Module 4 of Maintenance Engineering &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding and hydraulic system maintenance. Quality/process standardization using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding and hydraulic system maintenance. quality/process standardization should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 3 Understanding and hydraulic system maintenance. Quality/process standardization means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 3 Understanding and hydraulic system maintenance. Quality/process standardization □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ 3 Understanding methods & techniques. Trouble shooting & Importance of

3 Understanding methods & techniques. Trouble shooting & Importance of is an official topic in Module 4 of Maintenance Engineering &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding methods & techniques. Trouble shooting & Importance of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding methods & techniques. trouble shooting & importance of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 3 Understanding methods & techniques. Trouble shooting & Importance of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] 3 Understanding methods & techniques. Trouble shooting & Importance of [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▼ 3 Understanding calibrating printing machine (offset). Quality: ISO definition. Standardization - ISO definition. Need of quality control. Advantages of quality control/standardization. Quality assessment methods - subjective (visual), objective, measured. How a customer determine the quality. Customer satisfaction. Factors of influence and specifications determining the quality of the offset print. Introduction to Different process/quality/standard/material research firms/organization - ISO, GATF, FOGRA, UGRA, PIRA, Brunner, IGT, CIE, ICC, Printing Industries of America (PIA). Quality & process control targets / patches: GATF/Printing Industries Plate Control Target. UGRA/FOGRA media wedge. UGRA plate control target film. UGRA/FOGRA process control elements in. UGRA/FOGRA Digital plate control wedge. UGRA/FOGRA digital print scale. Color control Bars - parts of color control bar - tint patches, solid patches, grey patches, star targets, RGB overprints, total area coverage patches. GATF CTP printing plate exposure wedge. GATF Grey balance patch. GATF ICC color charts. GATF test forms - contains different targets and sizes. GATF star target. GATF line screen (visual and measured). Brunner zebra strip. Brunner color control strip. Brunner CTP plate control strip. Brunner CMYK marks. Registration patches for unit to unit registration. Quality/process standardization methods & techniques: Optimization. Lean printing definition, Advantages. Lean printing definition. Lean goals. 7 types of wastes in Lean. Lean tools - 5s, standardized work instructions, value stream mapping, Total proactive maintenance, Kaizen events, Just in time management, 5 whys, etc. Six Sigma - Six Sigma concept, definition, advantages. Troubleshooting and Importance of calibrating printing machine (offset). Text / Reference: T/R Book Title/Author T1 GATF Lithographers manual GATF , USA. R1 Herschel L Apfelberg Maintaining printing equipment GATF. R2 Barbera, L Albinini Solving web offset press problem GATF. R3 HP Garg Industrial maintenance S Chand and company ltd. Online resources: Sl. No Website Link 1 http://edelmann-printingmachines.com/ 2 https://learnmech.com › mechanical-drives-belt-chain-gear 3 http://www.yourarticlelibrary.com/industries/maintenance-management

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Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding calibrating printing machine (offset). Quality: ISO definition. Standardization - ISO definition. Need of quality control. Advantages of quality control/standardization. Quality assessment methods - subjective (visual), objective, measured. How a customer determine the quality. Customer satisfaction. Factors of influence and specifications determining the quality of the offset print. Introduction to Different process/quality/standard/material research firms/organization - ISO, GATF, FOGRA, UGRA, PIRA, Brunner, IGT, CIE, ICC, Printing Industries of America (PIA). Quality & process control targets / patches: GATF/Printing Industries Plate Control Target. UGRA/FOGRA media wedge. UGRA plate control target film. UGRA/FOGRA process control elements in. UGRA/FOGRA Digital plate control wedge. UGRA/FOGRA digital print scale. Color control Bars - parts of color control bar - tint patches, solid patches, grey patches, star targets, RGB overprints, total area coverage patches. GATF CTP printing plate exposure wedge. GATF Grey balance patch. GATF ICC color charts. GATF test forms - contains different targets and sizes. GATF star target. GATF line screen (visual and measured). Brunner zebra strip. Brunner color control strip. Brunner CTP plate control strip. Brunner CMYK marks. Registration patches for unit to unit registration. Quality/process standardization methods & techniques: Optimization. Lean printing definition, Advantages. Lean printing definition. Lean goals. 7 types of wastes in Lean. Lean tools - 5s, standardized work instructions, value stream mapping, Total proactive maintenance, Kaizen events, Just in time management, 5 whys, etc. Six Sigma - Six Sigma concept, definition, advantages. Troubleshooting and Importance of calibrating printing machine (offset). Text / Reference: T/R Book Title/Author T1 GATF Lithographers manual GATF , USA. R1 Herschel L Apfelberg Maintaining printing equipment GATF. R2 Barbera, L Albinini Solving web offset press problem GATF. R3 HP Garg Industrial maintenance S Chand and company ltd. Online resources: Sl. No Website Link 1 http://edelmann-printingmachines.com/ 2 https://learnmech.com › mechanical-drives-belt-chain-gear 3 http://www.yourarticlelibrary.com/industries/maintenance-management using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding calibrating printing machine (offset). quality: iso definition. standardization - iso definition. need of quality control. advantages of quality control/standardization. quality assessment methods - subjective (visual), objective, measured. how a customer determine the quality. customer satisfaction. factors of influence and specifications determining the quality of the offset print. introduction to different process/quality/standard/material research firms/organization - iso, gatf, fogra, ugra, pira, brunner, igt, cie, icc, printing industries of america (pia). quality & process control targets / patches: gatf/printing industries plate control target. ugra/fogra media wedge. ugra plate control target film. ugra/fogra process control elements in. ugra/fogra digital plate control wedge. ugra/fogra digital print scale. color control bars - parts of color control bar - tint patches, solid patches, grey patches, star targets, rgb overprints, total area coverage patches. gatf ctp printing plate exposure wedge. gatf grey balance patch. gatf icc color charts. gatf test forms - contains different targets and sizes. gatf star target. gatf line screen (visual and measured). brunner zebra strip. brunner color control strip. brunner ctp plate control strip. brunner cmyk marks. registration patches for unit to unit registration. quality/process standardization methods & techniques: optimization. lean printing definition, advantages. lean printing definition. lean goals. 7 types of wastes in lean. lean tools - 5s, standardized work instructions, value stream mapping, total proactive maintenance, kaizen events, just in time management, 5 whys, etc. six sigma - six sigma concept, definition, advantages. troubleshooting and importance of calibrating printing machine (offset). text / reference: t/r book title/author t1 gatf lithographers manual gatf , usa. r1 herschel l apfelberg maintaining printing equipment gatf. r2 barbera, l albinini solving web offset press problem gatf. r3 hp garg industrial maintenance s chand and company ltd. online resources: sl. no website link 1 http://edelmann-printingmachines.com/ 2 https://learnmech.com › mechanical-drives-belt-chain-gear 3 http://www.yourarticlelibrary.com/industries/maintenance-management should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding calibrating printing machine (offset). Quality: ISO definition. Standardization - ISO definition. Need of quality control. Advantages of quality control/standardization. Quality assessment methods - subjective (visual), objective, measured. How a customer determine the quality. Customer satisfaction. Factors of influence and specifications determining the quality of the offset print. Introduction to Different process/quality/standard/material research firms/organization - ISO, GATF, FOGRA, UGRA, PIRA, Brunner, IGT, CIE, ICC, Printing Industries of America (PIA). Quality & process control targets / patches: GATF/Printing Industries Plate Control Target. UGRA/FOGRA media wedge. UGRA plate control target film. UGRA/FOGRA process control elements in. UGRA/FOGRA Digital plate control wedge. UGRA/FOGRA digital print scale. Color control Bars - parts of color control bar - tint patches, solid patches, grey patches, star targets, RGB overprints, total area coverage patches. GATF CTP printing plate exposure wedge. GATF Grey balance patch. GATF ICC color charts. GATF test forms - contains different targets and sizes. GATF star target. GATF line screen (visual and measured). Brunner zebra strip. Brunner color control strip. Brunner CTP plate control strip. Brunner CMYK marks. Registration patches for unit to unit registration. Quality/process standardization methods & techniques: Optimization. Lean printing definition, Advantages. Lean printing definition. Lean goals. 7 types of wastes in Lean. Lean tools - 5s, standardized work instructions, value stream mapping, Total proactive maintenance, Kaizen events, Just in time management, 5 whys, etc. Six Sigma - Six Sigma concept, definition, advantages. Troubleshooting and

Aspect	What to prepare
	Importance of calibrating printing machine (offset). Text / Reference: T/R Book Title/Author T1 GATF Lithographers manual GATF , USA. R1 Herschel L Apfelberg Maintaining printing equipment GATF. R2 Barbera, L Albinini Solving web offset press problem GATF. R3 HP Garg Industrial maintenance S Chand and company ltd. Online resources: Sl. No Website Link 1 http://edelmann-printingmachines.com/ 2 https://learnmech.com › mechanical-drives-belt-chain-gear 3 http://www.yourarticlelibrary.com/industries/maintenance-management means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 3 Understanding calibrating printing machine (offset). Quality: ISO definition. Standardization - ISO definition. Need of quality control. Advantages of quality control/standardization. Quality assessment methods - subjective (visual), objective, measured. How a customer determine the quality. Customer satisfaction. Factors of influence and specifications determining the quality of the offset print. Introduction to Different process/quality/standard/material research firms/organization - ISO, GATF, FOGRA, UGRA, PIRA, Brunner, IGT, CIE, ICC, Printing Industries of America (PIA). Quality & process control targets / patches: GATF/Printing Industries Plate Control Target. UGRA/FOGRA media wedge. UGRA plate control target film. UGRA/FOGRA process control elements in. UGRA/FOGRA Digital plate control wedge. UGRA/FOGRA digital print scale. Color control Bars - parts of color control bar - tint patches, solid patches, grey patches, star targets, RGB overprints, total area coverage patches. GATF CTP printing plate exposure wedge. GATF Grey balance patch. GATF ICC color charts. GATF test forms - contains different targets and sizes. GATF star target. GATF line screen (visual and measured). Brunner zebra strip. Brunner color control strip. Brunner CTP plate control strip. Brunner CMYK marks. Registration patches for unit to unit registration. Quality/process standardization methods & techniques: Optimization. Lean printing definition, Advantages. Lean printing definition. Lean goals. 7 types of wastes in Lean. Lean tools - 5s, standardized work instructions, value stream mapping, Total proactive maintenance, Kaizen events, Just in time management, 5 whys, etc. Six Sigma - Six Sigma concept, definition, advantages. Troubleshooting and Importance of calibrating printing machine (offset). Text / Reference: T/R Book Title/Author T1 GATF Lithographers manual GATF , USA. R1 Herschel L Apfelberg Maintaining printing equipment GATF. R2 Barbera, L Albinini Solving web offset press problem GATF. R3 HP Garg Industrial maintenance S Chand and company ltd. Online resources: Sl. No Website Link 1 http://edelmann-printingmachines.com/ 2 https://learnmech.com › mechanical-drives-belt-chain-gear 3 http://www.yourarticlelibrary.com/industries/maintenance-management [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle →
Components/steps →
Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure
→ Observation → Result →
Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
methods - subjective
(visual), objective, measured. How a customer determine the quality. Customer
satisfaction.
Factors of influence and specifications determining the quality of the offset
print.
Introduction to Different process/quality/standard/material research
firms/organization -
ISO, GATF, FOGRA, UGRA, PIRA, Brunner, IGT, CIE, ICC, Printing Industries of
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registration. Quality/process standardization methods & techniques:
Optimization. Lean
printing definition, Advantages. Lean printing definition. Lean goals. 7 types
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

▶ **Define or explain Introduction to maintenance.. (3 marks)**

▶ **Define or explain Different types of maintenance. (3 marks)**

▶ **Define or explain Safety precautions and housekeeping. 4 Remembering Types of lubrications and lubricating. (3 marks)**

▶ **Define or explain 4 Understanding methods in printing industry Maintenance - Definition, Objectives and functions. Types of Maintenance - Planned maintenance and unplanned maintenance. Planned maintenance - Types-Preventive Maintenance, Predictive Maintenance, and Scheduled maintenance. Unplanned maintenance - Breakdown Maintenance. Contract maintenance. - Preventive Maintenance - Advantages, Functions - Planning, scheduling - Repair cycles, Dispatching and Controlling. Categories - Inspection, Small repair, Medium repair and Overhauling. Safety Precautions and House Keeping - safety precautions to be followed in press area - Aids to good housekeeping. Lubrication - Introduction - Advantages - Types of Lubricants - Solid, Semisolid and Liquid. Lubrication Schedule - Chart and Paint Marks. Methods of Lubrication - Gravity feed method-Wick feed, oil feed and Drip Feed , Force feed or Pressure feed method - Oil can, grease gun and grease cup, Splash method - Ring oiler, and Continuous/Multipoint flow method.. (3 marks)**

Module 2

▶ **Define or explain Chain drives.. (3 marks)**

▶ **Define or explain Belt drives.. (3 marks)**

▶ **Define or explain Gear drives.. (3 marks)**

▶ **Define or explain Bearings.. (3 marks)**

Module 3

▶ **Define or explain Machine erection.. (3 marks)**

▶ **Define or explain Machine reconditioning.. (3 marks)**

▶ **Define or explain Electrical drives. 5 Understanding Machine Erection and Reconditioning - Introduction, Equipments and Tools used for it, and application. Machine Erection - Equipment needed for erection - selection of location and environmental conditions - erection procedure for various prepress. Printing and finishing equipments and machinery. Reconditioning - Principles of reconditioning-repair methods of various parts - roller copperising and rerubberising - ebonite covering-dampening and inking systems - paper transport systems - cylinder bearing supports - eccentrics. Maintenance of electrical systems - AC motors and DC motors- electromagnetic friction couplings. Electromagnets - magnetic starters and contractors - limit switches - knife switches - micro switches starting and regulating rheostat - electric panels - electrical apparatus and electric wiring on the machines.. (3 marks)**

Module 4

▶ **Define or explain Quality. 1 Understanding Quality & Process control targets /. (3 marks)**

▶ **Define or explain 4 Understanding patches. Electrical system, pneumatic system. (3 marks)**

▶ **Define or explain 3 Understanding and hydraulic system maintenance. Quality/process standardization. (3 marks)**

▶ **Define or explain 3 Understanding methods & techniques. Trouble shooting & Importance of. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Introduction to maintenance.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Chain drives.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Machine erection.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram,

Module 4

1-mark definition: State the meaning of **Quality. 1 Understanding Quality & Process control targets /**.

3-mark explanation: Explain one principle, process or comparison from this module.

table, procedure, example or application.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl. No Website Link http://edelmann-printingmachines.com/
- https://learnmech.com › mechanical-drives-belt-chain-gear
- http://www.yourarticlelibrary.com/industries/maintenance-management

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 5103B. Verify institutional updates against the current official curriculum.